Mohith Pukale

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Experience

Principal Software Engineer — Identity & Access Infrastructure *Seattle, WA*
Oracle Cloud Infrastructure *Aug 2019 – present*

- Own identity and authorization infrastructure for **isolated, sovereign, and dedicated cloud environments**, spanning the internal authorization platform, just-in-time access service, and identity provider federation.
- Broke a cyclic dependency between the authorization platform and a legacy corporate identity provider, enabling **phishing-resistant hardware login** in place of OTPs; migrated affected users with no loss of access.
- Led **credential rotation automation** for a security hardening initiative; bootstrapped the rotator service, added service principals, and integrated an **SSH certificate authority** rotations take **under 5 min**.
- Built security-key programming and **PKI** certificate provisioning for multi-factor workstation login across Windows, Oracle Linux, and secure facilities, letting regional teams issue keys to **1,000+ users** locally.
- Designed an isolated administrative environment giving support engineers **scoped, audited access** to dedicated cloud distributions, and drove it through **architecture and security review boards**.
- Delivered **disaster recovery and multi-region** administrative login, keeping dedicated regions reachable during a regional failure, with per-region timeouts that adjust dynamically for degraded networks.
- Stood up a Windows Active Directory proof of concept end to end (**network, jump hosts, gateway**, directory, group-policy login), then designed the production architecture as a reusable deployment blueprint.
- Authored the **infrastructure-as-code provider** for the authorization service so partner teams declare access resources directly; cut resource-creation time **~50%** for the largest partner team.
- Cut entities under periodic access review from **~5,000 to ~200** added batching to the directory pipeline so a bulk change across **~40k accounts** landed incrementally instead of blocking on one long-running thread.

Software Developer — Consumer Knowledge Platform *Seattle, WA*
Amazon *Aug 2015 – Aug 2017*

- Re-architected the platform's core service with a new data model and metrics, giving **25% faster** response times.
- Reduced operational load **15%** by decomposing a monolithic service into a federated one.
- Implemented automated recovery for **big-data analytics workflows**, with failure and delay notification.

Software Developer Intern — Consumer Marketing Analytics *Seattle, WA*
Amazon *Jan 2015 – July 2015*

- Automated aggregation and publishing of customer scores with Apache Pig and AWS workflow services.

Software Developer Intern — QuickBooks Online *June 2014 – Aug 2014*
Intuit

- Built a service **masking sensitive production data** for QA; cut request time **20%** (MySQL to MongoDB).

Research

Graduate Student Researcher — Fault-Tolerant Oblivious Data Store *June 2018 – Aug 2018*
University of California, Santa Barbara

- Added **Oblivious RAM** to fault-tolerant cloud storage, hiding data access patterns from the storage provider.

Education

University of California, Santa Barbara *Santa Barbara, CA*
MS in Computer Science *Sept 2017 – June 2019*

- GPA: 3.9/4.0. Coursework: Adv. Security, Adv. Cryptography, Cloud Computing, Information Retrieval

PES Institute of Technology *Sept 2011 – June 2015*
BE in Information Science & Engineering

- GPA: 9.25/10. Coursework: Computer & Network Security, Data Mining

Technical Skills

Languages: Go, Java, Python, C, Bash, SQL, PowerShell

Security & Identity: OAuth 2.0, SAML, OIDC, LDAP, Active Directory, PKI, FIDO U2F, YubiKey, SSH CA, cryptography

Cloud & Networking: OCI, AWS, virtual cloud networks, Terraform-style IaC, disaster recovery, CI/CD, Linux, Windows

Data & Systems: Distributed systems, Elasticsearch, Solr, MySQL, MongoDB, Apache Hadoop, Apache Pig